UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. Which statement is most suitable for selecting among several constant cases?
   A. for
   B. do...while
   C. while
   D. switch

2. What is displayed by the following code?

```
int x = 7;
cout << x / 2 << " " << x % 2;
```

   A. 3 0
   B. 4 1
   C. 3 1
   D. 3.5 1

3. Which values can a bool variable hold?
   A. true or false
   B. any decimal
   C. any character
   D. any integer

4. How many times does a while loop execute when its condition is false initially?
   A. Until the user stops it
   B. Two
   C. One
   D. Zero

5. A computer system's hardware includes which of the following?
   i. input devices
   ii. output devices
   iii. central processing unit
   iv. main memory
   v. secondary storage
   A. i, ii, iii, iv
   B. i, ii, iii, iv, v
   C. i, ii, iii, v
   D. i, iii, iv

6. Which phase of the program development life cycle defines what the program must do?
   A. Testing
   B. Maintenance
   C. Analysis
   D. Coding

7. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. i

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B. t
C. p
D. n

8. Which declaration creates an integer array of five elements?
A. int a[5];
B. array a[5];
C. int a;
D. int a(5);

9. Which of the following are valid identifiers?
i. total_score
ii. 2total
iii. _count
iv. int
A. i and ii
B. i and iii
C. ii and iv
D. iii and iv

10. Which structure chooses between two alternative paths?
A. if...else
B. for
C. struct
D. array

PART B (60 MARKS)

QUESTION 1

This question tests high-level languages and debugging.

a) Define syntax error and logic error.

(2 marks)

b) Explain TWO advantages of high-level languages over machine language.

(3 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade shown in Table 1.

Table 1. Grade Scale

Score	Grade
80-100	A
60-79	B
50-59	C
40-49	D
0-39	F

a) Draw a flowchart for the grade classification.

(6 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(7 marks)

QUESTION 3

This question tests operator precedence and assignment.

a) Evaluate the following expressions using the declarations in the code block.

```
int a = 12, b = 5;
double c = 2.5;
```

(6 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

(5 marks)

QUESTION 4

Study the following C++ program segment:

```
int total = 0;
for (int i = 1; i <= 5; i++) {
    total += i;
}
cout << total;
```

a) Trace the program and state the output.

(7 marks)

b) Rewrite the loop using a while loop. Then explain what changes if the condition is i < 5.

(8 marks)

QUESTION 5

A shop stores the item prices shown in Table 1.

a) Declare an array named prices using the values in Table 1. Display all values.

Table 2. Item Prices

Item	Price (RM)
1	20.50
2	55.00
3	15.90
4	75.00
5	10.00

(8 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(8 marks)

PART C (30 MARKS)

QUESTION 1

A catering company charges the meal prices shown in Table 1.

Table 3. Meal Price

Quantity Range	Price per Meal (RM)
50 - 99	11.50
100 - 199	10.50
200 and above	9.50

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

(15 marks)

QUESTION 2

A cafe sells the items shown in Table 1. A 10% discount is given when the subtotal is at least RM50.

Table 4. Cafe Menu

Item	Price (RM)
Coffee	8.50
Tea	6.00
Juice	9.50

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

(15 marks)

END OF QUESTION PAPER

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